USN | | | | | | | | | | [24MTEEM101]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

FOUNDATION OF ENERGY ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	BL	Marks
1.	a.	Explain the concept of control volume and control surface in thermodynamics. How do they differ from the system approach? Provide examples to illustrate.	1	2	10
	b.	Define a thermodynamic process and thermodynamic cycle. Explain the differences between the two and give an example of each.	1	2	10
2.	a.	Define capillarity. Derive an expression for capillary rise of fluid	2	3	10
	b.	Explain the concepts of compressibility and bulk modulus. Derive the relationship between them	2	3	10
3.	a.	Explain the three modes of heat transfer. For each mode, describe the basic governing laws on the heat transfer mechanism.	3	2	10
	b.	Define thermal conductivity. Discuss the factors that affect the thermal conductivity of materials and explain how this property is significant in designing thermal insulation for buildings.	3	2	10
		OR			
4.	a.	Define the convective heat transfer coefficient. Discuss the factors that influence the convective heat transfer coefficient in both forced and natural convection scenarios.	3	2	10
	b.	Discuss the combined heat transfer mechanisms that occur in real-world situations. Provide examples of boundary conditions used in different thermal systems.	3	3	10
5.	a.	Discuss how power generated is transmitted from source (Pow plants) to the end user through grids using transmission lines	4	3	10
	b.	Explain with a neat sketch the construction and working principle of synchronous generator.	4	2	10
		OR			
6.	a.	Discuss the construction and working principle of induction motor.	4	3	10
	b.	Compare and contrast the working principles of synchronous and asynchronous generator.	4	4	10

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7.	**a.**	Explain the concept of energy and categorize energy resources.	**5**	**2**	**06**
	b.	Discuss the following concepts from worlds scenario i. Importance of energy consumption as a measure of energy prosperity ii. Energy science & technology	**5**	**3**	**14**
		OR			
8.	**a.**	Discuss the importance of non-conventional energy resources and analyze their role in meeting global energy demands.	**5**	**4**	**10**
	b.	What is geo thermal energy? Explain the issues associated in harnessing geothermal energy?	**5**	**2**	**10**